RV College of Engineering®

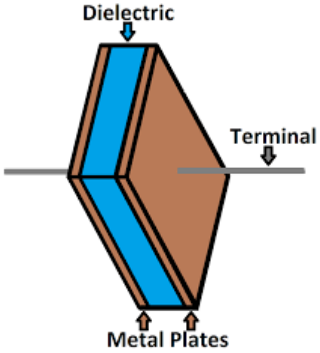
Unit-II
CAPACITOR

Go, change the world®

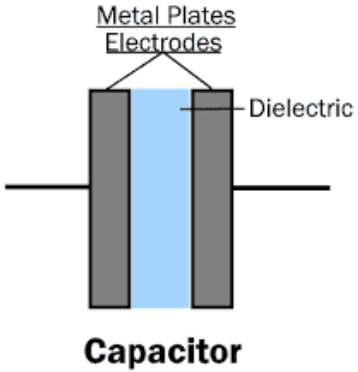
What Is a Capacitor?

A capacitor is a two-terminal electrical device that can store energy in the form of an electric charge

Capacitor is a device that is used to store an electric charge. It is basically an arrangement of conductors.



A capacitor typically consists of **an insulator with two metal plates attached on both sides of the insulator**. Insulators do not conduct current. The insulator used in capacitors is called a dielectric.



Difference between capacitor and battery

Capacitor





CAPACITOR

- The potential energy is stored in the electric field

BATTERY

- The potential energy is stored in the form of chemical energy which is later converted to electric energy



Capacitor

- ✗ Energy
- ✓ Weight
- ✓ Cost
- ✓ Charge speed
- ✓ Lifespan
- ✓ Materials



Battery

- ✓ Energy
- ✗ Weight
- ✗ Cost
- ✗ Charge speed
- ✗ Lifespan
- ✗ Materials

Difference between capacitor and battery

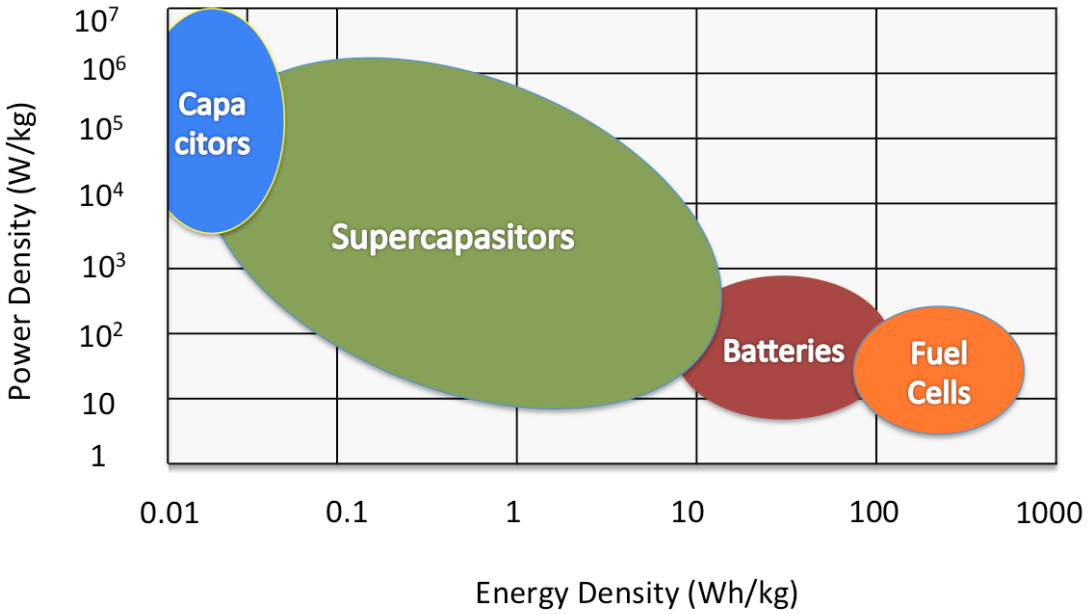
Capacitors:

- Higher power density
- Over a million charge-discharge cycles
- Environmentally friendly
- Extremely low internal resistance (reactance)
- High efficiency (97-98%)
- Much faster charge and discharge rate

Batteries:

- Have higher energy density
- Typically 200–1000 charge-discharge cycles
- Contain highly reactive and hazardous chemicals
- Negatively effected by low temperatures
- Slower and steady charge / discharge rate
-

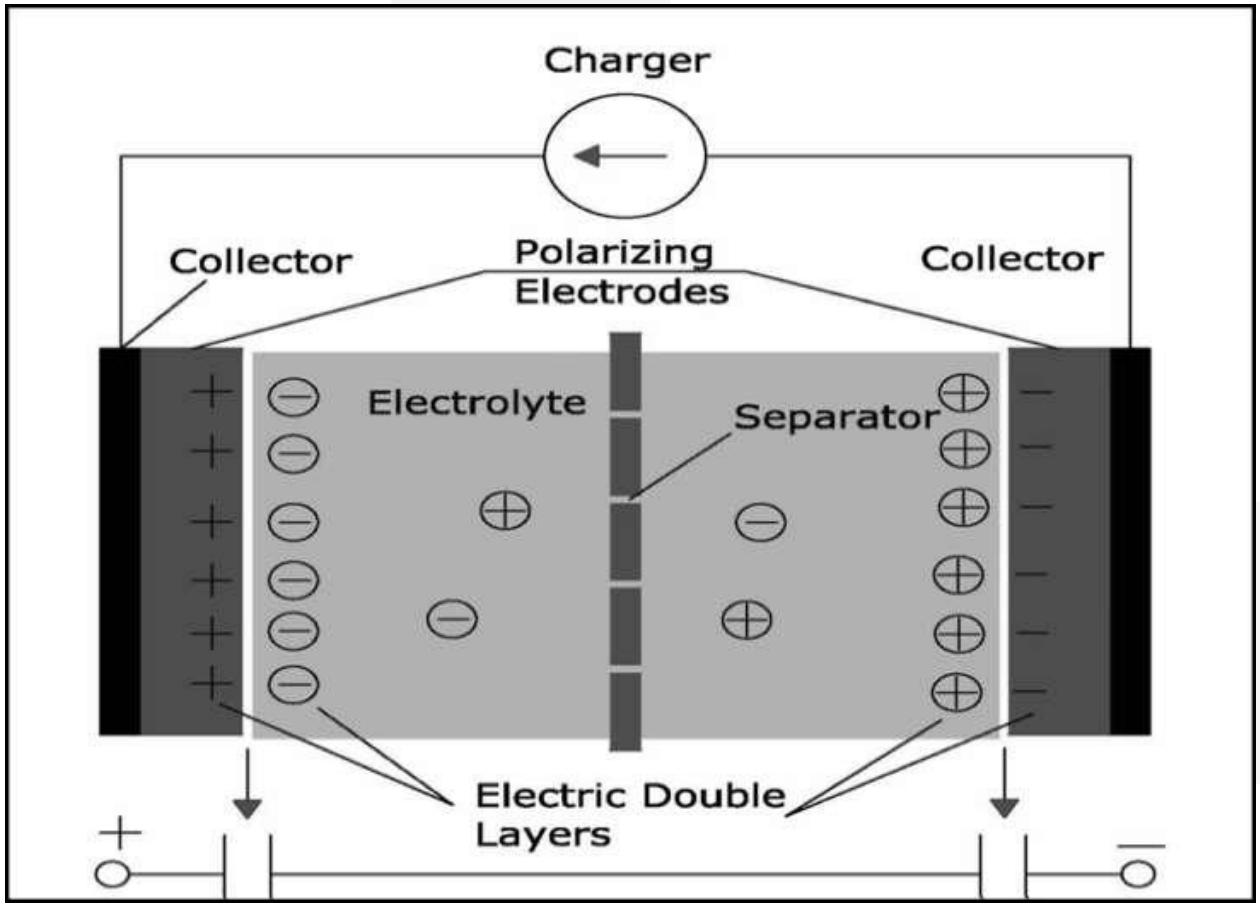
Capacitor



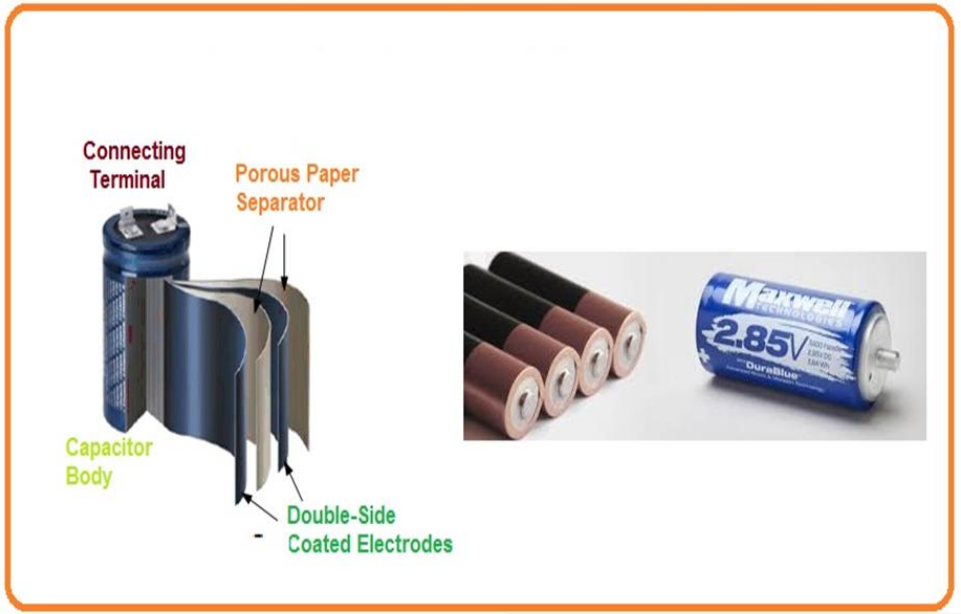
Power density refers to how quickly a device can discharge its energy, while energy density refers to how much energy a device contains.

Power density is ratio of power available from a battery to its mass (W/kg)
Energy density is defined as the energy available from a cell to its weight (Whr/kg)

Supercapacitors



Supercapacitors

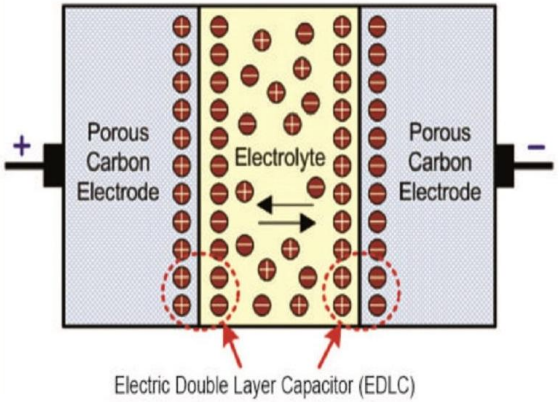


- A **supercapacitor** is an electrochemical capacitor that has an very high energy density as compared to common capacitors, about 100 times greater.
- Supercapacitor is also known as Electric Double Layer Capacitor(EDLC) or Ultracapacitor.
- The capacitance range is From 100 Farad to 5K Farad

Super capacitors

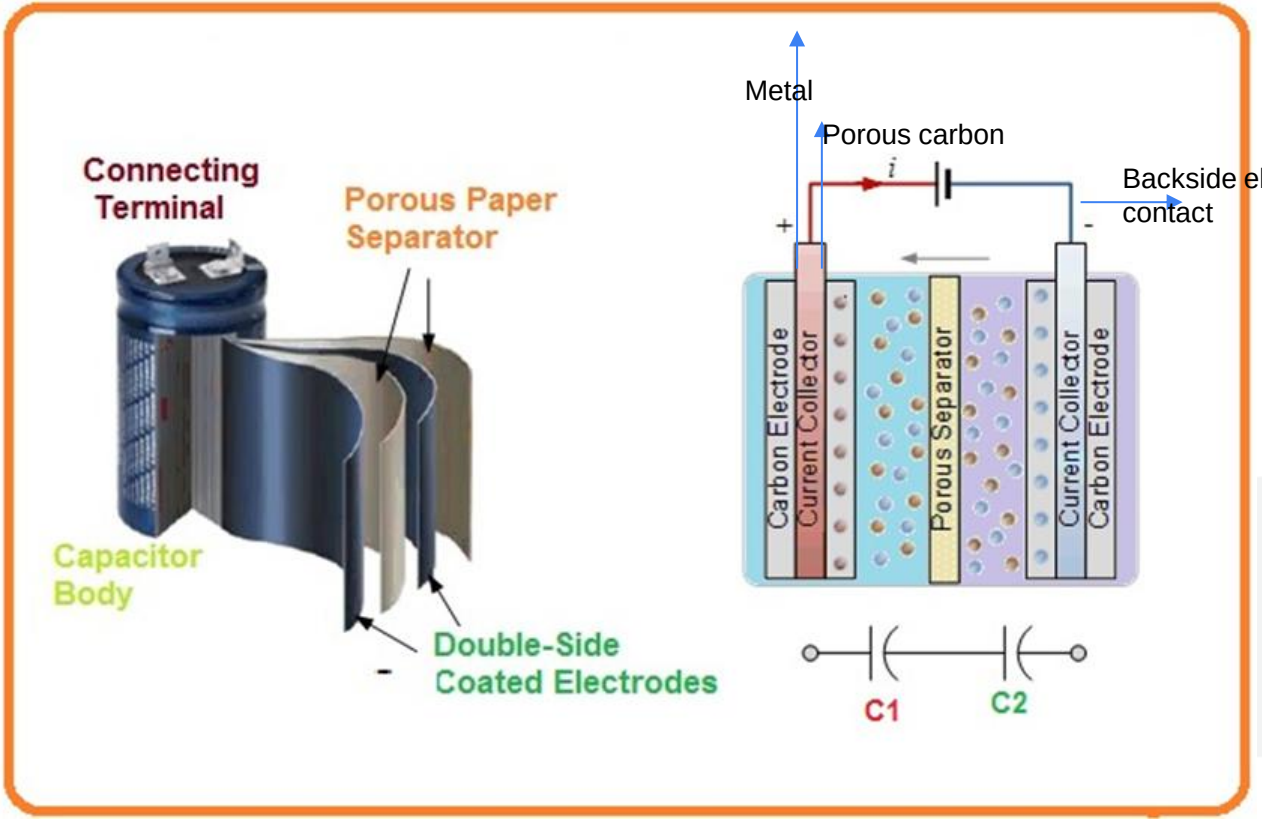
supercapacitors consist of two porous electrodes, electrolyte, a separator and current collectors.

(i) Electrode material
(ii) Electrolyte material
(iii) Current collector
(iv) Binder
(v) Separators

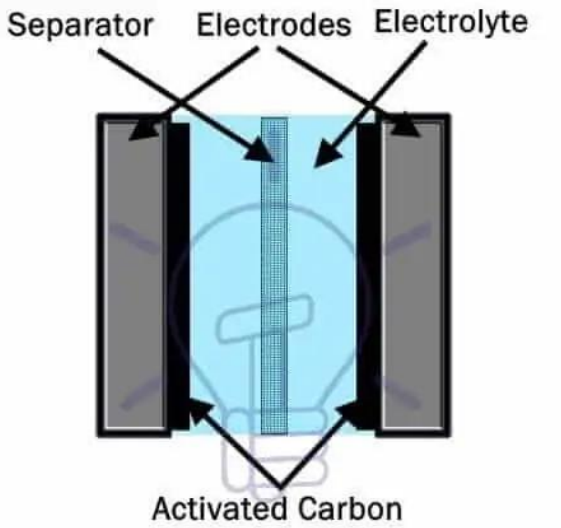


A **supercapacitor** or **ultracapacitor** is an electrochemical capacitor that has an unusually high energy density when compared to common capacitors. They are of particular interest in automotive applications for hybrid vehicles and as supplementary storage for battery electric vehicles.

Construction of super capacitor



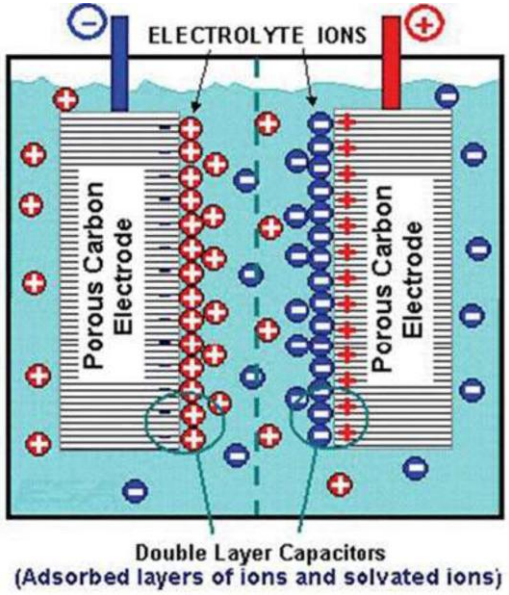
Super capacitor



SuperCapacitor Construction

Supercapacitors can safely and regularly discharge 100% of their stored energy, while batteries can only discharge **50-80%** of their stored energy

Supercapacitors are considered between Capacitors and Batteries. The main reason is that Supercapacitor charges/discharges very fast like a capacitor and its capacity is high.



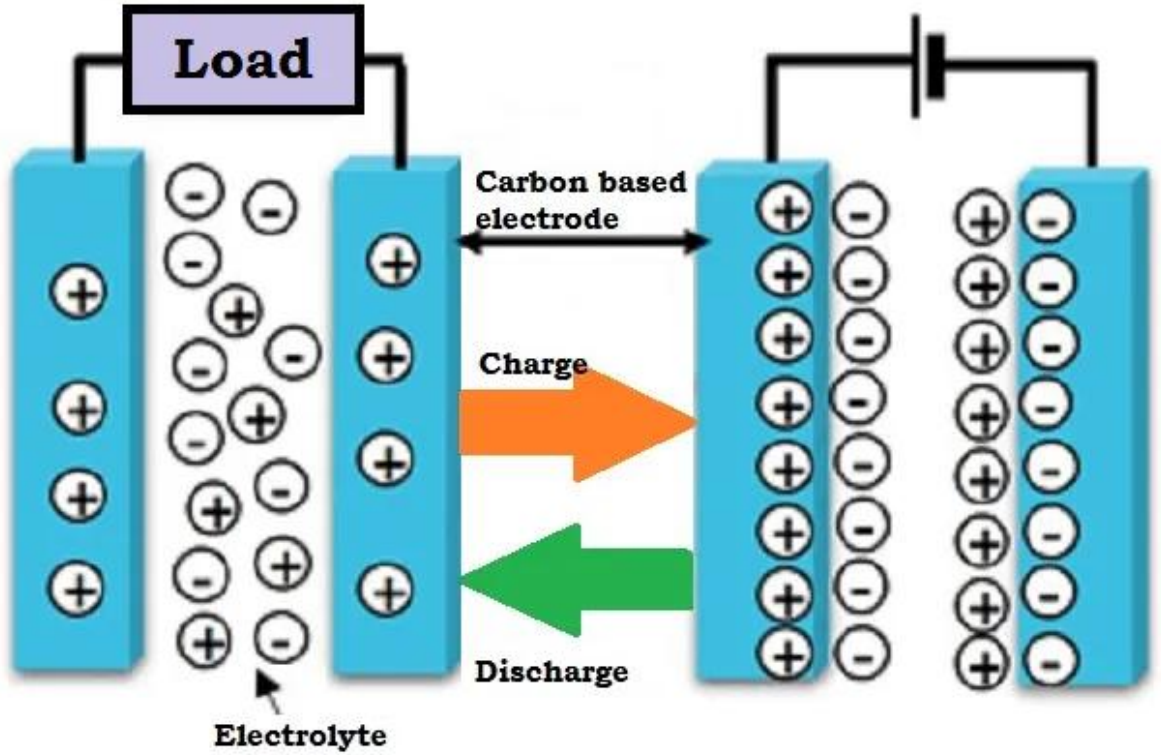
Differences between capacitor and supercapacitors

Capacitor	Supercapacitor
Capacitors are passive electrical devices with two terminals that retain charge as an electric field between their metal plates.	Supercapacitors are also known as double-layer capacitors, ultracapacitors, or super cap that can hold a huge amount of electric charge.
Capacitors consist of two metal plates serving as the anode and cathode of electrodes that are separated by a material known as the dielectric.	In supercapacitors, active carbon is used as the electrode material for electrodes. Supercapacitors use dielectric materials as a separator between the anode and cathode.
The current travel through the insulating material when the source voltage is supplied between the two terminals, but the material resists the flow of electrons.	Supercapacitors use either electrostatic double-layer capacitance (EDLC) or a combination of the two known as a hybrid capacitance to store charge.
Even though the voltage across the capacitor terminal is equal to the applied voltage, the insulation material continues to stop the flow of electrons. This resistance leads to a shift that has the effect of storing energy as an electrostatically charged field.	They are made up of electrodes with layers of activated carbor The separator is pressed in between them. An ion-permeable membrane such as graphene acts as the separator and allows for the exchange of electrolyte ions between the electrodes.

Capacitor vs Super capacitor

- The main difference that exists between a capacitor and a supercapacitor is that supercapacitors are based on carbon technology.
- Due to this carbon technology, these capacitors are able to create a very large surface area.
- This area exists even with an extremely small separation distance.

Working of Supercapacitors



Working of Supercapacitors

Positively and negatively charged ions are present between the two plates of the supercapacitor's electrolyte solution.

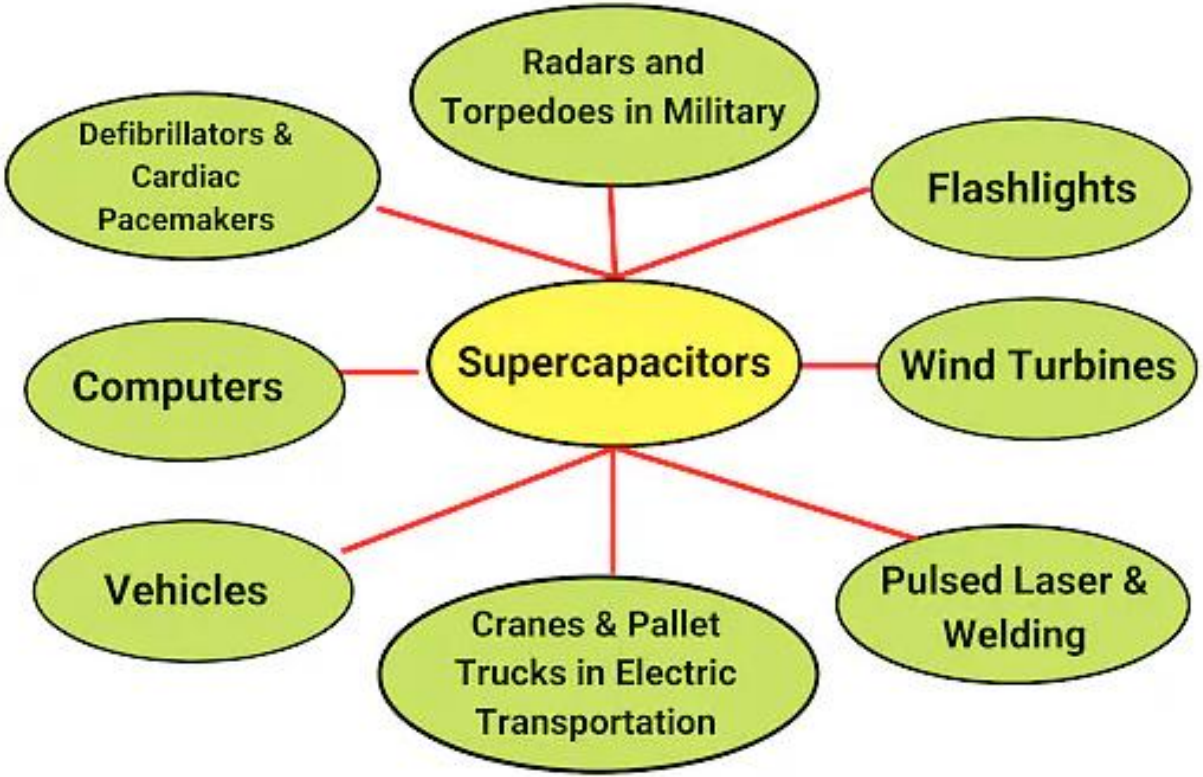
One of the plates develops a positive charge whereas the other plate develops a negative charge when a voltage is placed across the plates.

The positively charged plate attracts the negatively charged ions in the electrolyte solution, whereas the negatively charged metal plate attracts the positively charged ions towards itself.

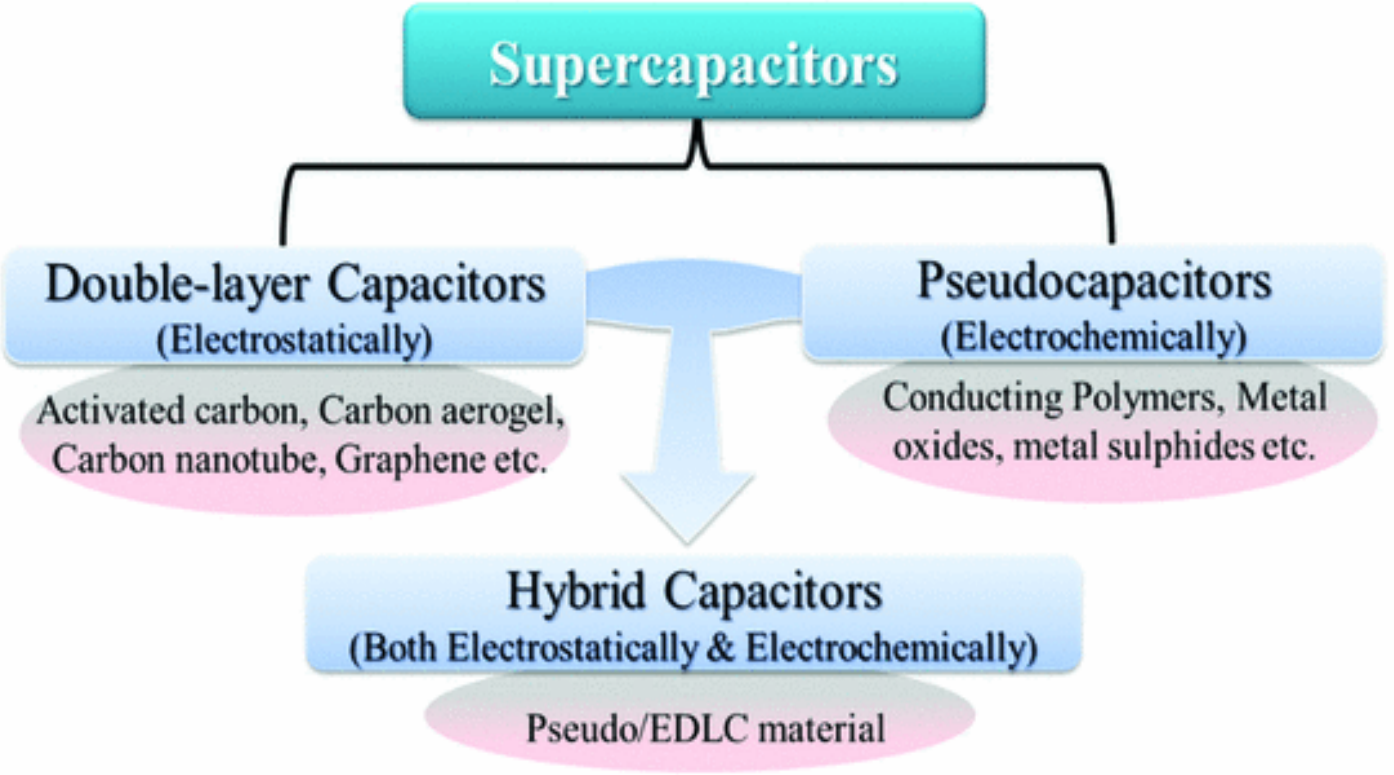
A thin coating of ions is deposited on the inner surface of both plates.

An electrostatic double layer is formed as a result, which is like connecting two capacitors in series.
Since the space between their charge layers is relatively small, each of the two resulting capacitors has a high capacitance value.
A supercapacitor's total capacitance can be calculated by dividing (C1 × C2) by (C1 + C1).

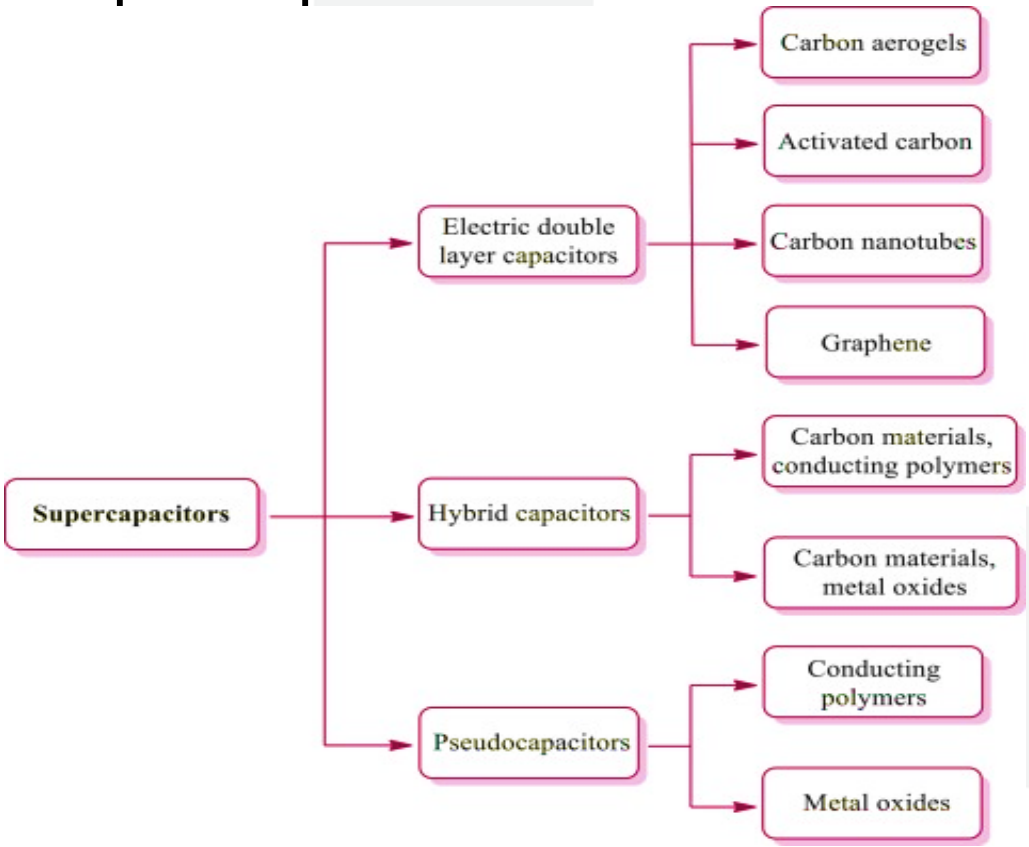
Applications of Supercapacitors



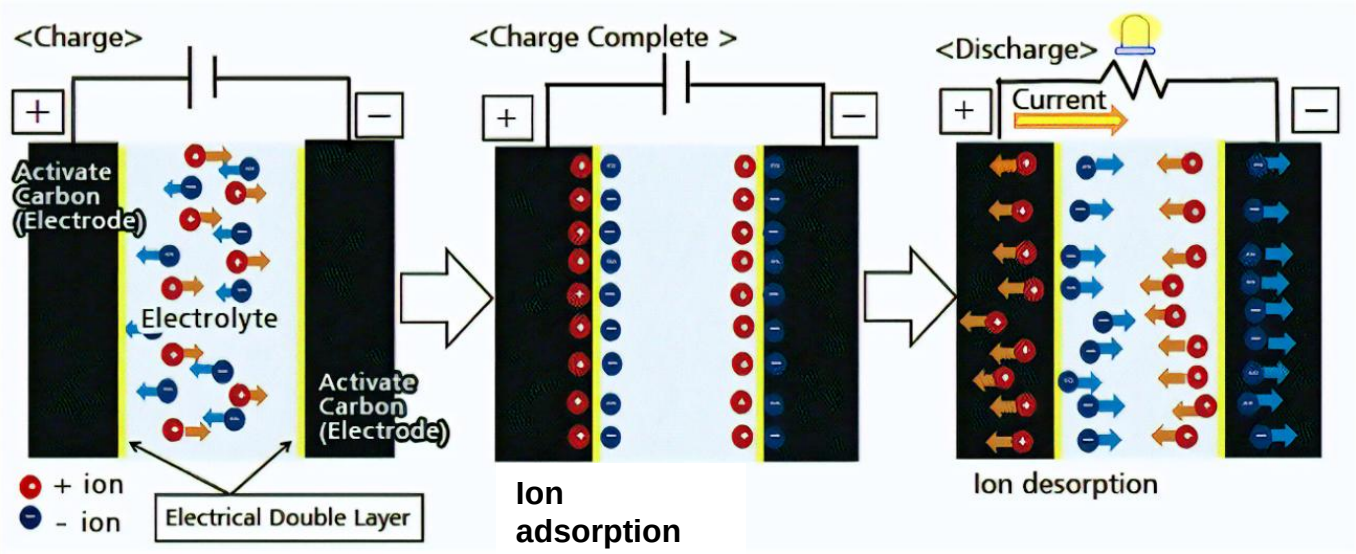
Classification of super capacitors



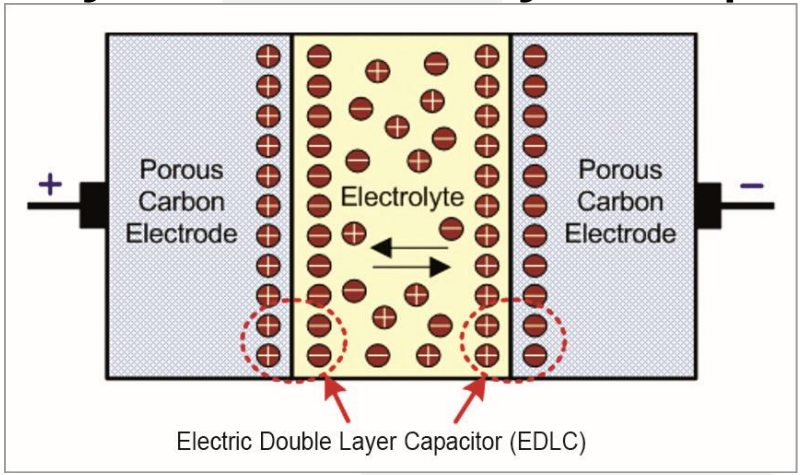
Super capacitors classification



Electrolytic double layer capacitor



Electrolytic double layer capacitor



•Electrostatic Double Layer Capacitors comprises two **electrodes**, a separator, and an electrolyte.

•Electrolyte is a **mixture** comprising positive and negative **ions** dissolved in water.

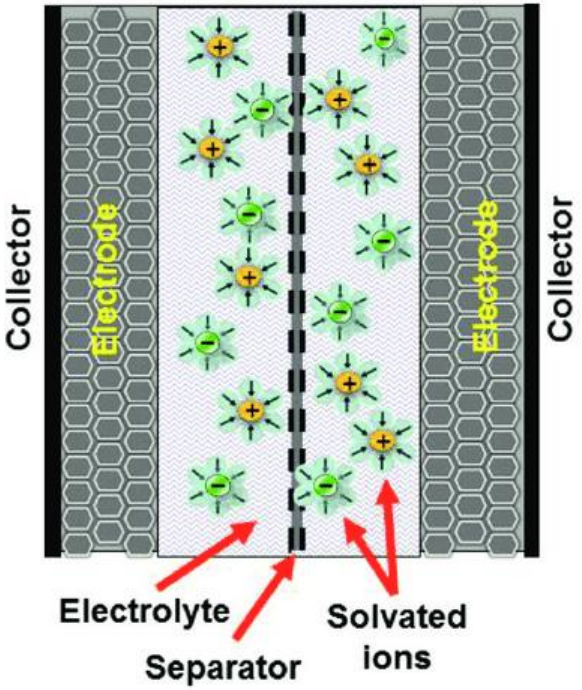
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•The two electrodes are separated from each other through a separator.

•The supercapacitors use carbon electrodes with much higher electrostatic double-layer capacitance.

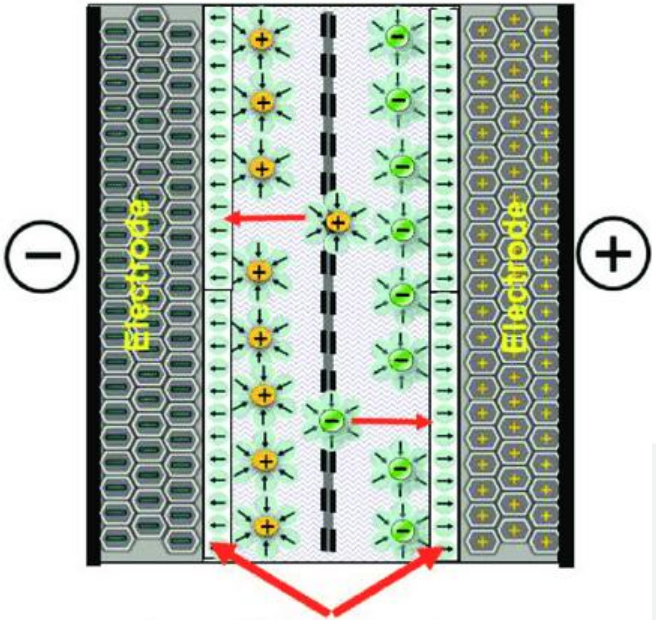
Capacitor discharged/charged

Capacitor discharged



Random distribution of ions

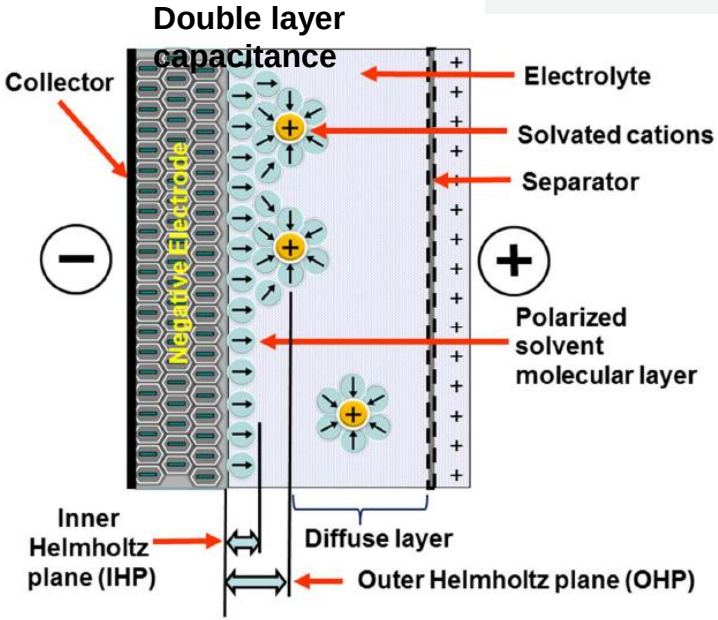
Capacitor charged

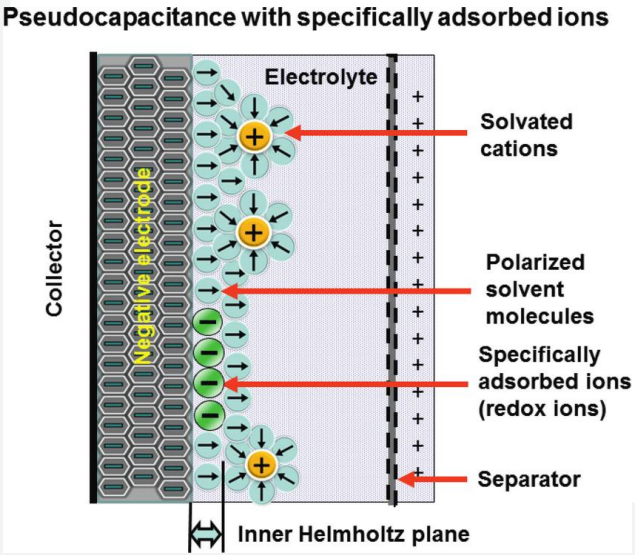


Inner Helmholtz plane
(polarized solvent molecules)

Mirror image of charge distribution of ions in opposite polarity

Pseudo capacitors



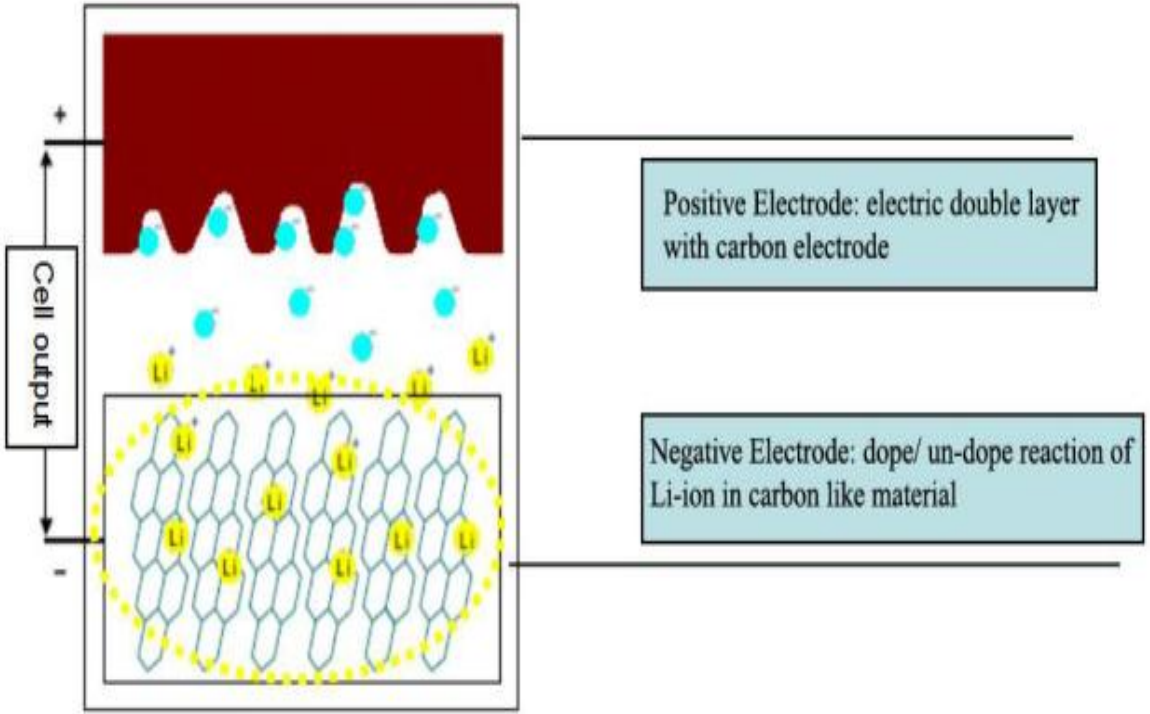


1. Inner Helmholtz potential
2. Outer Helmholtz potential
3. Specifically Adsorption of ion
4. Redox reaction(Specifically adsorbed ions-redox ions)

Pseudo Capacitors

- Pseudo Capacitors are also referred to as electrochemical pseudo-capacitors.
- They make use of metal oxide or conducting polymer electrodes that have a high amount of electrochemical pseudocapacitance.
- They store electrical energy by **electron** charge transfer between electrode and electrolyte.
- It is done through the **oxidation and reduction reaction** commonly known as a **redox reaction**.

Schematics representation of hybrid capacitor



Disadvantages of Supercapacitors

- Supercapacitors discharge themselves much more frequently which is significantly higher than a battery.
- There are low voltages that exist within individual cells due to which series connections are necessary to attain greater voltages.
- There is a significantly lesser amount of energy stored per unit weight in comparison to an electrochemical battery.
- Supercapacitors deliver a poor energy density compared to batteries. It is equal to between one-fifth and a tenth of the battery's energy.
- They cannot be utilized in circuits with AC or higher frequencies.